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NODE JS + EXPRESS + MONGODB CRUD API PRACTICAL
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STEP 1 : INSTALL NODE JS
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```

Download Node.js

<https://nodejs.org>

After install check:

```
node -v
```

```
npm -v
```

```
=====
STEP 2 : INSTALL MONGODB
=====
```

Download MongoDB Community Server

<https://www.mongodb.com/try/download/community>

While installing:

✓ Install MongoDB Compass

```
=====
STEP 3 : START MONGODB
=====
```

Open terminal and run:

```
mongod
```

Leave this terminal running.

```
=====
STEP 4 : CREATE PROJECT
=====
```

```
mkdir crud-api
```

```
cd crud-api
```

```
=====
STEP 5 : INITIALIZE PROJECT
=====
```

```
npm init -y
```

```
=====
STEP 6 : INSTALL PACKAGES
=====
```

```
npm install express mongoose body-parser
```

```
=====
STEP 7 : CREATE FILE
=====
```

```
Create file:
```

```
server.js
```

```
=====
STEP 8 : WRITE CODE
=====
```

```
PASTE THIS CODE INSIDE server.js
```

```
-----

const express = require('express');
const mongoose = require('mongoose');
const bodyParser = require('body-parser');
const app = express();

/* BODY PARSER */
app.use(bodyParser.json());

/* MONGODB CONNECTION */
mongoose.connect('mongodb://127.0.0.1:27017/studentDB')
.then(() => console.log("MongoDB Connected"));

/* SCHEMA */
const StudentSchema = new mongoose.Schema({
  name: String,
  course: String
```

```
});
```

```
/* MODEL */
```

```
const Student = mongoose.model('Student', StudentSchema);
```

```
/* CREATE API */
```

```
app.post('/addStudent', async(req,res)=>{
```

```
    const student = new Student({
```

```
        name:req.body.name,
```

```
        course:req.body.course
```

```
    });
```

```
    await student.save();
```

```
    res.send("Student Added");
```

```
});
```

```
/* READ API */
```

```
app.get('/students', async(req,res)=>{
```

```
    const data = await Student.find();
```

```
    res.json(data);
```

```
});
```

```
/* UPDATE API */
```

```
app.put('/updateStudent/:id', async(req,res)=>{
```

```
    await Student.findByIdAndUpdate(
```

```
        req.params.id,
```

```
        {
```

```
            name:req.body.name,
```

```
            course:req.body.course
```

```
        }
```

```

    );

    res.send("Student Updated");
  });

/* DELETE API */
app.delete('/deleteStudent/:id', async(req,res)=>{
    await Student.findByIdAndDelete(req.params.id);
    res.send("Student Deleted");
  });

/* SERVER */
app.listen(3000, ()=>{
    console.log("Server Running on Port 3000");
  });

```

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STEP 9 : RUN PROJECT

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Open NEW terminal:

node server.js

=====

STEP 10 : TEST APIs USING POSTMAN

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Install Postman:

<https://www.postman.com/downloads/>

=====

1) CREATE API

=====

METHOD:
POST

URL:

http://localhost:3000/addStudent

BODY → raw → JSON

```
{  
  "name": "Atharva",  
  "course": "BCA"  
}
```

=====

2) READ API

=====

METHOD:
GET

URL:

http://localhost:3000/students

=====

3) UPDATE API

=====

METHOD:
PUT

URL:

http://localhost:3000/updateStudent/id

Replace id with MongoDB id

BODY:

```
{  
  "name": "Rohit",  
  "course": "MCA"  
}
```

=====

4) DELETE API

=====

METHOD:
DELETE

URL:

http://localhost:3000/deleteStudent/id

Replace id with MongoDB id